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*Views expressed herein are my own and not those of my employer

LLMs

How to Solve Software Problems with LLMs

DON'T.

LLMs Are Weird

Two of my favorite properties of software:

- Reproducibility and Determinism

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Non-exhaustive list of things LLMs are expressly terrible at:

- Reproducibility and Determinism



LLMs Are Squishy

Squishy Problems
call for Squishy
Solutions

Example 1 - Summarize (gpt-4o-mini)

“Please summarize this article” + <text from
<https://simonwillison.net/2025/Apr/8/stop-syncing-everything/>>

The article discusses Graft, a new open-source Rust data synchronization engine developed by Carl Sverre, which he presented at the Vancouver Systems meetup. Graft organizes data into Volumes, which are sparse ordered sets of Pages, with a fixed size of 4KB. It maintains a complete history of each volume through snapshots, allowing clients to read and write from specific versions of pages while being updated on changes without needing to synchronize all changed data immediately.

Graft is especially suited for B-tree databases like SQLite and includes a SQLite VFS extension that implements multi-leader read-write replication on top of Graft volumes. Carl's presentation includes a demo and documentation for users to explore.

Example 2 - Tighten up (gpt-4o-mini)

“Please summarize this article in one or two sentences - the goal is to create a headline for a url bookmark entry” + <text from <https://simonwillison.net/2025/Apr/8/stop-syncing-everything/>>

Headline for Bookmark Entry:

"Introducing Graft: A Revolutionary
Open Source Rust Data
Synchronization Engine for Efficient
Data Management and Replication"

Example 3 - Deeper (gpt-4o-mini)

“Please suggest five content tags for a blog content tagging system based on the content of this text” + <text from <https://simonwillison.net/2025/Apr/8/stop-syncing-everything/>>

Here are five content tags for a blog content tagging system based on the provided text about Graft:

1. **Data Synchronization**
2. **Open Source**
3. **Rust Programming**
4. **SQLite Integration**
5. **Offline-First Applications**

These tags capture the key themes and concepts discussed in the text, making it easier for readers to locate relevant content.

Squishy Things LLMs are Good At

- Summarization
- Classification
- Data Enrichment
- Interpreting Intent
- Natural Language Processing
- Writing Language
- Writing Software

How to Reason About LLMs

- Like an API - make a request, get data
- But with “moods”
 - Wants to be helpful
 - Misunderstands the request
 - Refuses to answer
 - Breaks with no explanation
- Conversation partner, squishy bridge builder

Crash Course

- Prompt is the instruction + context + history
- It's your job to curate the conversation context to solve problems
 - Seeing behavior you don't like? Tell it to stop in the prompt
 - Output not quite right? Add example scenarios
 - Struggling with the prompt? Ask an LLM for help
 - Anthropic Prompt Improver

Crash Course

- Progressively transform your data from most to least squishy
- Experiment with chaining calls together
 - LLM based grading, formatting, decisions, looping, branching

Crash Course - Integration

- Structured output
 - A JSON specification for the LLM to match in output
 - Squishy Bridge A
- Tool use
 - Tell the LLM about cool APIs or functions you have available to manipulate data
 - Squishy Bridge B

How to Get Started - Practically

Use Your Brain

How to Get Started - Practically

Practice

How to Get Started - Practically

- **Single ecosystem** (OpenAI, Anthropic, Gemini)
 - Use the official Python library
- **Multi-Ecosystem**
 - Use a library to paper over the differences (like LiteLLM)
- **Get some API keys and experiment**
 - It's weirdly, unnervingly cheap
 - Please don't commit your API key
 - .env + .gitignore + dotenv
- **Testing with LLMs is Weird** - look into eval frameworks

Model Cheat Sheet

- **Cheap, Fast:** Gemini Flash-2.0 / Lite, OpenAI 4o-mini
- **Writing, Coding:** Anthropic Sonnet 3.5 or 3.7
- **Data Enrichment, Problem Solving:** OpenAI o3-mini
- **All-Around, Multi-Modal:** Gemini Pro 2.5
 - Proceed with caution for production (for now)

How to Keep Up

- Take a deep breath
- Find what works, and stick with it
- Don't worry about the bleeding edge - it's a free-for-all gold rush
- Read everything written by these two gentlefolk:

Ethan Mollick

Wharton Gen AI Labs

<https://www.oneusefulthing.org/>

Simon Willison

Django, Datasette

<https://simonwillison.net/>

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Thanks to Dan S for the slide review

<https://ai-analytics.wharton.upenn.edu/generative-ai-labs/about-us/#:~:text=shapiro>